

安徽大学 2022-2023 学年第二学期

《电磁场与电磁波》期中考试试卷参考答案及评分标准

一. 填空题（每空 3 分，共 30 分）

1. $\vec{e}_x 5 - \vec{e}_y 3 - \vec{e}_z$

2. $2x\vec{e}_x + 4y\vec{e}_y + 6z\vec{e}_z$

3. $75\pi^2$

4. $112/5$

5. $-\frac{k}{r^2}, \quad \frac{k}{a}$

6. -10

7. $0, \quad \vec{e}_z 2a$

8. $\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$

二. 简答题（共 10 分）

9.
$$\begin{cases} \nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}, \text{表明传导电流和变化的电场都能产生磁场} \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \text{表明变化的磁场产生电场} \\ \nabla \cdot \vec{B} = 0, \text{表明磁场是无源场, 磁感线总是闭合曲线} \\ \nabla \cdot \vec{D} = \rho, \text{表明电荷产生电场, 是电场的散度源} \end{cases}$$

三. 证明题（共 12 分）

10.

证 (1)
$$\nabla \cdot \mathbf{R} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 3 \quad (4 \text{ 分})$$

(2)
$$\nabla \times \mathbf{R} = \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = 0 \quad (4 \text{ 分})$$

(3) 设 $\mathbf{A} = e_x A_x + e_y A_y + e_z A_z$, 则 $\mathbf{A} \cdot \mathbf{R} = A_x x + A_y y + A_z z$, 故

$$\begin{aligned}\nabla(\mathbf{A} \cdot \mathbf{R}) &= e_x \frac{\partial}{\partial x}(A_x x + A_y y + A_z z) + e_y \frac{\partial}{\partial y}(A_x x + A_y y + A_z z) + \\ &e_z \frac{\partial}{\partial z}(A_x x + A_y y + A_z z) = e_x A_x + e_y A_y + e_z A_z\end{aligned}\quad (4 \text{ 分})$$

四. 计算题 (共 48 分)

11. (本题 10 分)解: 设电流为 I , 由安培环路定理: $\oint \vec{H} \cdot d\vec{l} = I$ (2 分)

(1) 当所取回路半径 $0 \leq \rho \leq a$, 有

$$\vec{H} \cdot 2\pi\rho = \frac{I \cdot \pi\rho^2}{\pi a^2} \quad (2 \text{ 分})$$

$$\text{得到 } \vec{H} = \vec{e}_\phi \frac{I\rho}{2\pi a^2} \quad (2 \text{ 分})$$

(2) 当 $a \leq \rho \leq b$, 有

$$\vec{H} \cdot 2\pi\rho = I \quad (2 \text{ 分})$$

$$\text{得到 } \vec{H} = \vec{e}_\phi \frac{I}{2\pi\rho} \quad (2 \text{ 分})$$

12. (本题 10 分)

解: 由安培环路定理求出平行双线中得电流在矩形回路平面任意一点产生得磁感应强度分别为:

$$\begin{aligned}\vec{B}_{\text{左}} &= \frac{\mu_0 i}{2\pi r} \\ \vec{B}_{\text{右}} &= \frac{\mu_0 i}{2\pi(b+c+d-r)}\end{aligned}\quad (3 \text{ 分})$$

方向均垂直于纸面向内

$$\text{回路中的感应电动势 } \varepsilon_{\text{in}} = -\frac{d}{dt} \int_s \vec{B} \cdot d\vec{S} = -\frac{d}{dt} \left[\int_s \vec{B}_{\text{左}} \cdot d\vec{S} + \int_s \vec{B}_{\text{右}} \cdot d\vec{S} \right] \quad (2 \text{ 分})$$

$$\begin{aligned}\text{其中 } \int_s \vec{B}_{\text{左}} \cdot d\vec{S} &= \int_b^{b+c} \frac{\mu_0 i}{2\pi r} a dr = \frac{\mu_0 ai}{2\pi} \ln\left(\frac{b+c}{b}\right) \\ \int_s \vec{B}_{\text{右}} \cdot d\vec{S} &= \int_c^{d+c} \frac{\mu_0 i}{2\pi(b+c+d-r)} a dr = \frac{\mu_0 ai}{2\pi} \ln\left(\frac{b+c}{b}\right)\end{aligned}\quad (2 \text{ 分})$$

$$\begin{aligned}\varepsilon_{\text{in}} &= -2 \frac{d}{dt} \left[\frac{\mu_0 a i}{2\pi} \ln\left(\frac{b+c}{b}\right) \right] = -\frac{\mu_0 a}{2\pi} \ln\left(\frac{b+c}{b}\right) \frac{d}{dt} [0.1 \cos(2\pi \times 10^7 t)] \\ \text{则} &= \frac{4\pi \times 10^{-7} \times 0.5}{\pi} \ln 2 \times 0.1 \sin(2\pi \times 10^7 t) \times 2\pi \times 10^7 \\ &= 0.4\pi \times \ln 2 \sin(2\pi \times 10^7 t) \text{ V}\end{aligned}\quad (3 \text{ 分})$$

13. (本题 14 分)

解: (1) 由 $\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t}$

得

$$-e_x \frac{\partial E}{\partial z} + e_z \frac{\partial E}{\partial x} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t}$$

即

$$\begin{aligned}\frac{\partial \mathbf{H}}{\partial t} &= -\frac{E_0}{\mu_0} \left[-e_x \frac{\pi}{d} \cos\left(\frac{\pi z}{d}\right) \cos(\omega t - k_x x) \right. \\ &\quad \left. + e_z k_x \sin\left(\frac{\pi z}{d}\right) \sin(\omega t - k_x x) \right]\end{aligned}\quad (3 \text{ 分})$$

将上式对时间 t 积分, 得

$$\begin{aligned}\mathbf{H}(x, z, t) &= -\frac{E_0}{\mu_0} \left[-e_x \int \frac{\pi}{d} \cos\left(\frac{\pi z}{d}\right) \cos(\omega t - k_x x) dt + \right. \\ &\quad \left. e_z \int k_x \sin\left(\frac{\pi z}{d}\right) \sin(\omega t - k_x x) dt \right] \text{ A/m} \\ &= \left[e_x \frac{\pi E_0}{\omega \mu_0 d} \cos\left(\frac{\pi z}{d}\right) \sin(\omega t - k_x x) \right. \\ &\quad \left. + e_z \frac{k_x E_0}{\omega \mu_0} \sin\left(\frac{\pi z}{d}\right) \cos(\omega t - k_x x) \right] \text{ A/m}\end{aligned}\quad (3 \text{ 分})$$

(2) 面电流和面电荷出现在两个理想导体板的内表面上, 分别为:

在 $z=0$ 处的导体板上

$$\mathbf{J}_S = \mathbf{e}_n \times \mathbf{H}|_{z=0} = e_z \times \mathbf{H}|_{z=0} = \left[e_y \frac{\pi E_0}{\omega \mu_0 d} \sin(\omega t - k_x x) \right] \text{ A/m} \quad (2 \text{ 分})$$

$$\rho_S = \mathbf{e}_n \cdot \mathbf{D}|_{z=0} = e_z \cdot \varepsilon_0 \mathbf{E}|_{z=0} = 0 \quad (2 \text{ 分})$$

在 $z=d$ 处的导体板上

$$\mathbf{J}_S = \mathbf{e}_n \times \mathbf{H}|_{z=d} = -e_z \times \mathbf{H}|_{z=d} = \left[e_y \frac{\pi E_0}{\omega \mu_0 d} \sin(\omega t - k_x x) \right] \text{ A/m} \quad (2 \text{ 分})$$

$$\rho_S = \mathbf{e}_n \cdot \mathbf{D}|_{z=d} = -e_z \cdot \varepsilon_0 \mathbf{E}|_{z=d} = 0 \quad (2 \text{ 分})$$

14. (本题 14 分)

(1)

解：只有满足麦克斯韦方程组的矢量才可能是电磁场的场矢量。因此,利用麦克斯韦方程组确定 $\boldsymbol{E}$ 可能是电磁场的电场强度矢量的条件。

由式(2.6.6),得

$$\begin{aligned}\frac{\partial \boldsymbol{B}}{\partial t} &= -\nabla \times \boldsymbol{E} = -\begin{vmatrix} \boldsymbol{e}_x & \boldsymbol{e}_y & \boldsymbol{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = -\boldsymbol{e}_y \frac{\partial E_x}{\partial z} \\ &= -\boldsymbol{e}_y \frac{\partial}{\partial z} [E_m \cos(\omega t - kz)] = -\boldsymbol{e}_y k E_m \sin(\omega t - kz)\end{aligned}$$

对上式积分,得

$$\boldsymbol{B} = \boldsymbol{e}_y \frac{k E_m}{\omega} \cos(\omega t - kz) \quad (4 \text{ 分})$$

由 $\boldsymbol{B} = \mu \boldsymbol{H}$,得

$$\boldsymbol{H} = \boldsymbol{e}_y \frac{k E_m}{\mu \omega} \cos(\omega t - kz) \quad (2 \text{ 分})$$

由 $\boldsymbol{D} = \epsilon \boldsymbol{E}$,得

$$\boldsymbol{D} = \boldsymbol{e}_x \epsilon E_m \cos(\omega t - kz) \quad (2 \text{ 分})$$

$$\nabla \times \boldsymbol{H} = \begin{vmatrix} \boldsymbol{e}_x & \boldsymbol{e}_y & \boldsymbol{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = -\boldsymbol{e}_x \frac{\partial H_y}{\partial z}$$

$$= -\boldsymbol{e}_x \frac{k^2 E_m}{\omega \mu} \sin(\omega t - kz) \quad (2 \text{ 分})$$

而

$$\frac{\partial \boldsymbol{D}}{\partial t} = \boldsymbol{e}_x \frac{\partial D_x}{\partial t} = -\boldsymbol{e}_x \epsilon E_m \omega \sin(\omega t - kz) \quad (2 \text{ 分})$$

故

$$k^2 = \omega^2 \mu \epsilon$$

即

$$k = \pm \omega \sqrt{\mu \epsilon} \quad (2 \text{ 分})$$

(2)